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Note on Dr Glaisher's Paper on a Theorem in Definite Integration.

[From the Quarterly Journal of Pure and Applied Mathematics, vol. x. (1870), pp. 355, 356.]

IT is worth noticing that under the case when $\phi = 1$ may be proved independently of the general formula with Θ_1 for (1) the equation

$$s = ax - \frac{a_1}{a - b_1}x - \frac{a_2}{a - b_2}x - \dots - \frac{a_n}{a - b_n}x \dots,$$

is $(sx - r)(sx - b_1)(sx - b_2) \dots (sx - b_n) \dots = 0$, and has $s + 1$ roots, say $x_1, x_2, \dots, x_{n+1}$, where

$$x_1 = x, \quad x_2 = a - b_1 + b_1, \quad \dots, \quad x_{n+1} = a - b_n + b_n;$$

and (2) the equation

$$r = -\frac{a_1}{a - b_1}x - \frac{a_2}{a - b_2}x - \dots - \frac{a_n}{a - b_n}x \dots,$$

is $r(x - b_1)(x - b_2) \dots (x - b_n) \dots = 0$, and has n roots $x_1, x_2, \dots, x_n$ where

$$x_1 = x_1 - a_1, \quad x_2 = b_1 + b_2, \quad \dots, \quad x_n = \frac{x_{n-1} + b_{n-1} + b_n}{a_n};$$

whence

$$f dx_1 + f dx_2 + \dots + f dx_{n+1} = 0 \quad \text{in the first case}$$

and

$$nf(a, b)(x_n, x_{n-1}, \dots, x_2, x_1)' = 0 \quad \text{in the second}$$

[which are the two formulæ in question]

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On the Quartic Surfaces $(\ast U, V, W)^2 = 0$.

[From the *Quarterly Journal of Pure and Applied Mathematics*, vol. X. (1871), pp. 15–25.]

AMONG the surfaces of the form in question are included the reciprocals of several interesting surfaces of the orders 6, 8, 9, 10, and 12, viz.

Order 6, parabolic ring,

8, elliptic ring,

9, centro-surface of a paraboloid,

10, parallel surface of a paraboloid,

envelope of planes through the points of an ellipsoid at right angles to the radius vectors from the centre,

12, centro-surface of an ellipsoid,

parallel surface of an ellipsoid.

I propose to consider these surfaces, not at present in any detail, but merely for the purpose of presenting them in connexion with each other and with the present theory. It will be convenient to use homogeneous equations, but for the surcinal interpretation the coordinate W or w may be considered as equal to unity; I have not thought it necessary to so adjust the constants that the equations shall be homogeneous in regard to the constants; this case of course be done without difficulty, and in many cases it would be analytically advantageous to make the change.

I take throughout (X, Y, Z, W) for the coordinates of a point on the quartic surface (so that (X, Y, W) is, the equation $(\ast U, V, W)^2 = 0$ are to be considered as quartic functions of (X, Y, Z, W)); reserving (x, y, z, w) for the coordinates of a point on the reciprocal surface of the order 6, 8, 9, 10, or 12. The reciprocation is performed in regard to the imaginary sphere $x^2 + y^2 + z^2 + w^2 = 0$; the relation between

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the coordinates (X, Y, Z, W) and (x, y, z, w) is then $Xx + Yy + Zz + Ww = 0$, and the equation $(X, Y, Z, W) = 0$ is the equation in point-coordinates of the quartic surface, or in line-coordinates of the reciprocal surface; and similarly the equation $(x, y, z, w)^n = 0$ is the equation in point-coordinates of the reciprocal surface, or in line-coordinates of the quartic surface.

Parabolic ring, or envelope of a sphere of constant radius having its centre on a parabola.

Taking k for the radius of the sphere, and $x = 0$, $y^2 = ax$ for the equations of the parabola, then the coordinates of a point on the parabola are $a\theta^2$, $2a\theta$, 0 , where θ is a variable parameter. The equation of the sphere therefore is

$$(x - a\theta w)^2 + (y - 2a\theta w)^2 + z^2 - k^2 w^2 = 0,$$

and the ring is the envelope of this sphere.

The reciprocal of the sphere is

$$k^2(X^2 + Y^2 + Z^2) - (a\theta^2 X + 2a\theta Y + W)^2 = 0;$$

writing this in the form

$$a\theta^2X + 2a\theta Y + W = k\sqrt{1}(X^2 + Y^2 + Z^2) = 0,$$

and taking the envelope in regard to θ , we have

$$X\{W + k\sqrt{1}(X^2 + Y^2 + Z^2)\} - aY^2 = 0,$$

or, what is the same thing,

$$(aY^2 - XW)^2 = k^2X^2(X^2 + Y^2 + Z^2)$$

for the equation of the quartic surface. This has the form $X = 0, V = 0$ for a tangential line, but I am not in possession of a theory enabling me thence to infer that the parabolic ring is of the order 4.

To show that it is so, I have to find (x, y, z, w) which satisfy the equation of the variable sphere

$$(x - a\theta w)^2 + (y - 2a\theta w)^2 + z^2 - k^2w^2 = 0,$$

or, what is the same thing,

$$(A, B, C, D, E)(\theta, 1)^4 = 0,$$

where

$$A = a^2w^2,$$

$$B = 0,$$

$$C = a(2aw^2 - ax),$$

$$D = -4ayw,$$

$$E = x^2 + y^2 + z^2 - k^2w^2.$$

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Then $I = 3a^4w^2E$, $J = a^6w^6C$, and the equation is $I^3 - J^2 = 0$, viz. this is

$$(Ax^2 + 3y^2 - 3aw^2x + 3y^2 - 4xy)^3 - \{(3aw - x)(3y^2 + 3y^2 - 6wx - 4ayx) - 27a^2y^2wx\}^2(Ew - xw)^2 = 0,$$

or, as this may also be written,

$$(Ax^2 + 3y^2 - 3aw^2x)^3 - \{(3aw - x)(3y^2 + 3y^2 - 6wx - 4ayx) - 27a^2y^2wx\}^2(Ew - xw)^2 = 0.$$

Developing, the whole divides by 27, and the equation of the ring finally is

$$\begin{aligned} (y^2 - 6awx)^3(y^2 + xw) = & -3aw^2\{2x^2 - 2axw + 3aw^2(y - xw)(y^2 - 2yw)\}(z^2 + w^2) \\ & + \{3y^2 + y^2(2x - 2a)w + 24a^2(zw^2 + aw^2z) + 16a^2w^4(z^2 - xw)(z^2 - 8axw + 4a^2w^2)\}(z^2 - kw^2) \\ & + \{(2y^2 + axw(z^2 - kw^2))^2\}(z^2 - kw^2)^3 \end{aligned}$$

which is a quartic surface having for nodal conics

$$(y^2 - 6awx = 0, \quad z^2 + w^2 = 0), \quad (Z = 0, \quad x = w = 0),$$

the line at infinity being a torsal line.

Elliptic ring, or envelope of a sphere of constant radius having its centre on an ellipse.

Taking k for the radius of the sphere, and $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ for the equations of the ellipse, the coordinates of a point on the ellipse are $a \cos \theta$, $b \sin \theta$, 0; hence the equation of the variable sphere is

$$(x - aw \cos \theta)^2 + (y - bw \sin \theta)^2 + z^2 - k^2w^2 = 0.$$

The reciprocal of the sphere is

$$k^2(X^2 + Y^2 + Z^2) - (aX \cos \theta + bY \sin \theta + W)^2 = 0;$$

viz. writing this under the form

$$aX \cos \theta + bY \sin \theta + W = k\sqrt{(X^2 + Y^2 + Z^2)},$$

and taking the envelope in regard to θ , we have

$$\sqrt{(a^2X^2 + b^2Y^2)} = W + k\sqrt{(X^2 + Y^2 + Z^2)};$$

viz. this is

$$(a^2X^2 + b^2Y^2) + W^2 + k^2(X^2 + Y^2 + Z^2) - 2Wk\sqrt{(X^2 + Y^2 + Z^2)} = 0;$$

or

$$\{(a^2 + k^2)X^2 + (b^2 + k^2)Y^2 + k^2Z^2 + W^2\}^2 = 4W^2k^2(X^2 + Y^2 + Z^2),$$

for the equation of the quartic surface. This has the form $X = 0$, $V = 0$ for a tangential line, but I am not in possession of a theory enabling me thence to infer that the elliptic ring is of the order 4.

To show that it is so, I have to find (x, y, z, w) which satisfy the equation of the variable sphere

$$(x - aw \cos \theta)^2 + (y - bw \sin \theta)^2 + z^2 - k^2w^2 = 0,$$

or, what is the same thing,

$$(A, B, C, D, E)(t, 1)^4 = 0,$$

where

$$A = 0,$$

$$B = 0,$$

$$C = a(2aw^2 - ax),$$

$$D = -4byw,$$

$$E = 2(a^2 + b^2)w^2 + 2(x^2 + y^2 + z^2 - k^2w^2).$$

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For this purpose reverting to the equation

$$(x - aw \cos \theta + (y - bw \sin \theta + z^2 - k^2 w^2) = 0,$$

this may be written

$$A \cos 2\theta + B \sin 2\theta + C \cos \theta + D \sin \theta + E = 0,$$

where

$$\begin{aligned} A &= \frac{1}{2}(a^2 - b^2)w^2, \\ B &= 0, \\ C &= -4axw, \\ D &= -4byw, \\ E &= \frac{1}{2}(a^2 + b^2)w^2 + 2(x^2 + y^2 + z^2 - k^2 w^2), \end{aligned}$$

and the equation is

$$\begin{aligned} &12A(A - B) - 3(C^2 + D^2) + 4E^2 \\ &= \{12A^2(C^2 - D^2) - 24BCD - 3(C^2 + D^2)^2 + 4E^2(E - 8A^2)\}^2; \end{aligned}$$

or say

$$\{12A^2 + 3(C^2 + D^2) + 4E^2\}^3 - 27\{A(C^2 - D^2) - 2BCD - 3E(C^2 + D^2) + 8AE^2\}^2 = 0.$$

This is of the order 12, but it is easy to see that the terms in A^2 and $E^2(C^2 + D^2)$ disappear from the equation, all the other terms divide by w^2 ; and the equation is thus of the order 8.

The equation may be obtained somewhat differently, as follows. The equation of the variable sphere is

$$(x - aw)^2 + (y - bw)^2 + z^2 - k^2 w^2 = 0,$$

where (a, β) may subject to the condition $\frac{a^2}{a^2} + \frac{b^2}{b^2} = 1$. We have therefore

$$\frac{x - aw}{(x - aw) da} + \frac{y - bw}{(y - bw) db} = 0,$$

$$\beta da = \frac{a a}{a^2 + b^2} = -\frac{b}{a} da;$$

and thence

$$\begin{aligned} aw &= \frac{a^2 x}{a^2 + b^2}, & x - aw &= \frac{b^2 x}{a^2 + b^2}, \\ \beta w &= \frac{b^2 y}{a^2 + b^2}, & y - bw &= \frac{a^2 y}{a^2 + b^2}. \end{aligned}$$

Consequently

$$\begin{aligned} \frac{a^2 x^2}{(a^2 + b^2)^2} + \frac{b^2 y^2}{(a^2 + b^2)^2} - w^2 &= 0, \\ \frac{a^2 x^2}{(a^2 + b^2)^2} + \frac{b^2 y^2}{(a^2 + b^2)^2} - w^2 &= 0, \end{aligned}$$

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The quartic surface is consequently the envelope of the sphere

$$\frac{(a+b)^2}{a}x^2 + \frac{(b+c)^2}{b}y^2 + 2bz^2 - 2aw^2 = 0,$$

viz. this is

$$b\left(\frac{1}{a} - \frac{1}{a}\right)x^2 + 2\theta(x^2 + y^2 + z^2) + 2aX - 2bW = 0.$$

Hence the quartic surface is

$$\left(\frac{x^2}{a} - \frac{y^2}{a}\right)(aX^2 + bY^2 - 2ZW) = (X^2 + Y^2 + Z^2) = 0,$$

or, what is the same thing,

$$X^2Y^2(ab + cb) - 2ZW(bX^2 + aY^2) - 2abZ(X^2 + Y^2 + Z^2) - aBcZ = 0.$$

This has four nodes; viz. considering the equations

$$\frac{X}{a} + \frac{Y}{b} = 0, \quad aX^2 + bY^2 - 2ZW = 0, \quad X^2 + Y^2 + Z^2 = 0,$$

these give the quad. $X = 0, Y = 0, Z = 0$ four times, and four other points which are the nodes in question; the point $(X = 0, Y = 0, Z = 0)$ is a singular point of a higher order; the deduction caused by these singularities should be $= 8 + 19$, so as to make the order of the surface of curves of $36 - 8 - 19 = 12$, but it is the reduction on account of the point $(X = 0, Y = 0, Z = 0)$ must be $= 19$; but it is not by any means obvious how this is so.

Parallel surface of the paraboloid.

This is given, Salmon's *Solid Geometry*, 2nd Edit., pp. 146 and 148, [Ed. 4, p. 180], for the paraboloid $aX^2 + bY^2 + 2ZW = 0$, as the envelope of the quartic surface

$$\frac{a}{aw+1}X^2 + \frac{b}{bw+1}Y^2 + 2W(\theta - r^2w + r^2)w^2 = 0.$$

The reciprocal quartic is thus the envelope of

$$\frac{1}{a}x^2 + \frac{1}{b}y^2 + \frac{1}{a^2}z^2 + \frac{1}{a+r^2w}w^2 - w^2 = 0,$$

that is,

$$(X^2 + Y^2 + Z^2) + \frac{1}{b}\left(\frac{X^2}{a} - \frac{Y^2}{b} - \frac{2Z}{c}W\right) = 0;$$

whence the equation is

$$\frac{1}{b^2}(X^2 - Y^2)\left(\frac{X^2}{a} - \frac{Y^2}{b} - \frac{2Z}{c}\right) = 0,$$

viz. this is a quartic having the nodal lines

$$Z = X = 0, \quad Z = Y = 0$$

the circle must be equal, and the distance of the centre from the line of centres, and if the radii are r, r' , is readily found to be

$$(a - x)^2 + 2(r + r'),$$

suppose in general, that spheres any two contain the point P in the intersection of the points of P in regard to the two given conics respectively; then if P describes

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Envelope of the planes through the points of an ellipsoid at right angles to the radius vectors from the centre.

This is given in my paper “Note on a Surface,” in the *Annales de Matématique*, t. I. (1859), [206], as the envelope of the quartic surface

$$\frac{x^2}{1+\theta a} + \frac{y^2}{1+\theta b} + \frac{z^2}{1+\theta c} - \theta a^2 = 0.$$

The reciprocal quartic is thus the envelope of

$$\left(2 - \frac{x^2}{a}\right) X^2 + \left(1 - \frac{y^2}{b}\right) Y^2 + \left(1 - \frac{z^2}{c}\right) Z^2 - \frac{1}{a} W^2 = 0,$$

or, what is the same thing,

$$\theta \left(\frac{X^2}{a} + \frac{Y^2}{b} + \frac{Z^2}{c} - \frac{1}{c} W^2 \right) - (X^2 + Y^2 + Z^2) = 0;$$

viz. it is

$$W^2 = \frac{X^2 - aY^2}{X^2 - aY^2 + Z^2} = \frac{Y^2}{X^2 + Y^2 + Z^2} = \frac{Z}{X^2 + Y^2 + Z^2} = 0 \quad \text{for } X, Y, Z,$$

of the original $\frac{X^2}{a} + \frac{Y^2}{b} + \frac{Z^2}{c} = 1$; this is obvious geometrically inasmuch as the reciprocal of the variable plane is the inverse of the point on the ellipsoid.

The quartic surface has the nodal conic

$$W = 0, \quad aX^2 + bY^2 + cZ^2 = 0;$$

and also the node $X = 0, Y = 0, Z = 0$; there is a reduction in the order of the reciprocal surface a reduction $24 + 2 = 26$, or the order of the reciprocal surface is $= 10$.

Centro-surface of the ellipsoid.

Writing the equation of the ellipsoid in the form $\frac{x^2}{a} + \frac{y^2}{b} + \frac{z^2}{c} - w^2 = 0$, the centro-surface is given as the envelope of the quartic surface

$$\frac{x^2\theta^2}{(a+\theta)^2} + \frac{y^2\theta^2}{(b+\theta)^2} + \frac{z^2\theta^2}{(c+\theta)^2} - w^2 = 0.$$

(Salmon, [Ed. 2], p. 106; [Ed. 4, p. 177]), and hence the reciprocal quartic surface is the envelope of

$$\left(a + \frac{a^2}{\theta}\right) X^2 + \left(b + \frac{b^2}{\theta}\right) Y^2 + \left(c + \frac{c^2}{\theta}\right) Z^2 - W^2 = 0,$$

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in regard to the variable parameter θ , viz. the equation is

$$\left(\frac{X^2}{a} + \frac{Y^2}{b} + \frac{Z^2}{c}\right)(aX^2 + bY^2 + cZ^2 - W^2) - (X^2 + Y^2 + Z^2) = 0,$$

(see Salmon, [Ed. 2], p. 144 [Ed. 4, p. 172]). It hence at once appears, that the quartic surface has 12 nodes, viz. these are the four angles of the fundamental tetrahedron ($XYZW$), and the eight points

$$\begin{aligned}\frac{X^2}{a} + \frac{Y^2}{b} + \frac{Z^2}{c} &= 0, \\ X^2 + Y^2 + Z^2 &= 0, \quad W = 0, \\ aX^2 + bY^2 + cZ^2 - W^2 &= 0;\end{aligned}$$

or writing as is convenient to do,

$$(a, \beta, \gamma) = (b - c, c - a, a - b), \quad \alpha + \beta + \gamma = 0,$$

these are the eight points

$$(a, \beta, \gamma : \xi : \eta : \zeta) = 0,$$

the order of the reciprocal of the quartic surface is then $36 - 12 = 24$, which is in fact the order of the surface of centres.

The equation of the centro-surface is given, Salmon, [Ed. 2], p. 121, and *Quart. Math. Jour.*, t. IX. (1858), p. 220, in the form

$$(\theta, \beta, \gamma : \xi, \eta : \zeta)^4 = 0,$$

where ξ, η, ζ stand for ax, by, cz, bz ; it is therefore of the degree 18 in regard to (x, y, z, w) , which is the equation of the surface, but it must contain in itself the equation w^4 , and the order of the curve will be $20 - 6 = 14$.

Parallel surface of the ellipsoid.

This is given, Salmon, [Ed. 2], p. 148 [Ed. 4, p. 178], as the envelope of the quartic surface

$$\frac{a^2 X^2}{a^2 - r^2 w^2} + \frac{b^2 Y^2}{b^2 - r^2 w^2} + \frac{c^2 Z^2}{c^2 - r^2 w^2} = \left(1 - \frac{W^2}{w^2}\right) w^2.$$

The reciprocal quartic is thus the envelope of

$$(a^2 \theta X^2 + b^2 \theta Y^2 + c^2 \theta Z^2 + (a^2 - r^2) W^2 - W^2 + r^2 W^2)^2 = 0,$$

or writing $D + r^2 = \theta$, this is

$$(a^2 - 4\theta + b^2 + c^2 \theta) - (\theta - a^2) Y^2 - (\theta - c^2) Z^2 - (\theta - r^2) W^2 - W^2 - \left(1 - \frac{r^2}{\theta}\right) W^2 = 0.$$

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or, what is the same thing,

$$bX(bX - Z) + \{w^2 - r^2\theta + ay\}\{x^2 + (a+b)Y^2 + (a+c)Z^2 + (a-r^2)W^2 - W^2\} = 0,$$

whence the equation is

$$\{(a^2 - bY)(X^2 + (a-b)Y^2 + (a-c)Z^2 - W^2) + 2bW^2(X^2 + Y^2 + Z^2) = 0.$$

viz. this is a quartic having the nodal

$$W = 0, \quad (a^2 - bY)X^2 + (a-b)Y^2 + (a-c)Z^2 = 0.$$

The order of the reciprocal or parallel surface is then $36 - 24, = 12$, as it should be. The nodal conic of the quartic surface is the reciprocal of a tangent cone of the ellipsoid having a non-cyclic quadric cone, vertex the centre, in the parallel surface; this cone is imaginary for the ellipsoid, but real for either of the hyperboloids, and the existence in the case of the hyperboloid is readily perceived.

Reverting to the equation

$$\frac{a^2}{a^2 + \theta}X^2 + \frac{b^2}{b^2 + \theta}Y^2 + \frac{c^2}{c^2 + \theta}Z^2 - \left(1 + \frac{r^2W^2}{\theta}\right)w^2 = 0,$$

or say

$$(a^2 - P)X^2 + (b^2 - P)Y^2 + (c^2 - P)Z^2 - r^2(a^2 - P)(b^2 - P)(c^2 - P) = 0$$

this is

$$-w^2(a^2 - P)(b^2 - P)(c^2 - P) - r^2(a^2 - P)(b^2 - P)(c^2 - P) - P(a^2 - P)(b^2 - P)(c^2 - P) = 0,$$

where putting for shortness

$$\begin{aligned} u &= a^2 + b^2 + c^2, \\ \beta &= b^2c^2 + c^2a^2 + a^2b^2, \\ \gamma &= a^2b^2c^2, \\ b &= aP^2 - b^2, \\ p &= a^2 + y^2 + z^2, \\ q &= b^2c^2x^2 + a^2c^2y^2 + a^2b^2z^2, \\ r &= a^2b^2c^2w^2 - r^2a^2b^2c^2 + a^2b^2c^2, \end{aligned}$$

we have

$$\begin{aligned} A &= 12n^2, \\ B &= 9aw^2 - 2p, \\ C &= 12n - 4ap, \\ D &= 3aw - 3r, \\ E &= 12r^2. \end{aligned}$$

The equation of the parallel surface is

$$(AE - 4BD + 3C^2)^3 = 27(ACE - AD^2 - B^2E + 2BCD - C^3)^2.$$

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It is remarked (Salmon, [Ed. 2], p. 148 [Ed. 4, p. 176]) that there is in the plane $z = 0$, a nodal conic $\frac{x^2}{a^2+r^2} + \frac{y^2}{b^2+r^2} - 1 = 0$, the complete section being made up of this conic twice, and of the curve of the eighth order which is the parallel curve of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} - 1 = 0$; the like is of course the case as to the sections by the other two principal planes $x = 0$, $y = 0$. For the section by the plane $w = 0$ (or plane infinity) we have at once $p^2 + (qw^2 = 0$, where observe that

$$\begin{aligned} q^2 - 4pr &= \{(a^2 + b^2 + c^2)x^2 + (b^2c^2 + c^2a^2 + a^2b^2)y^2 + \dots\}^2 \\ &= (1, 1, 1, -1, -1, -1)(b^2 + c^2)y^2z^2, (c^2 - a^2)^2z^2x^2, (a^2 - b^2)^2x^2y^2) \\ &= ac(x^2 - y^2 - z^2 + b^2 + c^2(x^2 - y^2))^2, \end{aligned}$$

which has reduced to a sum of squares with positive signs.

The section is thus made up of two conics, each twice, and of four right lines: viz. the conics are $z^2 + y^2 + z^2 = 0$, this section at infinity of the sphere, the section at infinity of the ellipsoid; and the lines are

$$(\sqrt{X^2 - aZ^2} = 0, \quad (\sqrt{X^2 - aZ^2} + \sqrt{X^2 - bZ^2}, X^2 - aZ^2 = 0)$$

viz. those are the common tangents of the two conics. The circle at infinity is a nodal conic on the surface, which has thus 4 nodal conics.

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Note on a Relation between Two Circles.*[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 82, 83.]*

CONSIDER any two circles O, Q ; and let AC, BD, AB, BC be the common tangents touching the circles in the points $A, A_1, B, B_1, C, C_1, D, D_1$: the locus of a point P such that the pairs of tangents from it to the two circles respectively form a harmonic pencil, is a conic (locus of the points $A, A_1, B, B_1, C, C_1, D, D_1$); but this conic may break up into two lines, viz. if (as in the figure) the points A, B, D, D are in a line, then the points C, C', A', B' will be in a symmetrically situated line, and the conic breaks up into this pair of lines, meeting suppose in K . The condition

[Figure: Two circles with common tangents and centre line — diagram omitted]

for this, if a, c' are the distances of the centres from a fixed point in the line of centres, and if the radii are r, r' , is readily found to be

$$(a - c')^2 = 2(r^2 + r'^2).$$

Suppose in general, that lines any two contain the point P in the intersection of the points of P in regard to the two given conics respectively; then, if P describes

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a line, the locus of P is a conic passing through the three conjugate points of the given conic, if, however, the line which is the locus of P pass through one of the conjugate points, then the conic the locus of P' breaks up into a pair of lines: one of them a fixed line through the other two conjugate points; that is, the locus of P' is through the demonstrated conjugate points. That is, if the locus of P be a line through a conjugate point, the locus of P' is a line through the same conjugate point; but in every other case the locus of P' is a conic.

Reverting to the figure of the two circles, in order that it may be possible that the two lines AD and BC may be loci of points P , P' , related as above, it is necessary that K shall be a conjugate point of the two circles; that is, if the two circles intersect in points A , X lying symmetrically in the line of centres, then K must be the anti-points of A , X ; what is the same thing the chord AX shall be one of the same mentioned relation. And it then appears that given two circles, the necessary and sufficient conditions for the coexistence of the properties mentioned in the theorem are that they shall be equal circles touching each other.

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On the Porism of the In-and-Circumscribed Polygon, and the (2, 2) Correspondence of Points on a Conic.*[From the Quarterly Journal of Pure and Applied Mathematics, vol. XI. (1871), pp. 83–91.]*

THE present paper includes, as will at once be seen, much that is perfectly well known; but the separate theories regarded it seemed to me, to be put together; and there are, particularly as regards the unsymmetrical case afterwards referred to, some results which I believe to be new.

The porism of the in-and-circumscribed polygon has its foundation in the theory of the symmetrical (2, 2) correspondence of points on a conic; viz. a (2, 2) correspondence is such that to any given position of either point there correspond two positions of the other point; and is a symmetrical (2, 2) correspondence either point indifferently may be considered as the first point and the other of them will then be the second point of the correspondence. Or, what is the same thing, if x, y are the parameters which serve to determine the two points, then x, y are connected by an equation of the form $(\alpha x, 1\alpha y, 1)^2 = 0$; or, as it will always thus, whatever the parameters (x, y) . In the case of such symmetrical relation it is easy to show that the line joining the two points envelopes a conic. For the relation may be expressed in the form $(\alpha x + y, x + y, xy)^2 = 0$; we may imagine the coordinates (P, Q, R) of each manner that for the point (x) on the first conic the corresponding values (P, Q, R) for and for the point (y) , $P : Q : R = 1 : y : y^2$; the equation of the line joining the two points is then

$$\begin{vmatrix} P & Q & R \\ 1 & x & x^2 \\ 1 & y & y^2 \end{vmatrix} = 0,$$

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that is

$$Pxy - Q(x + y) + R = 0,$$

or representing this by the equation

$$PL + QM + R = 0,$$

we have $L : -2 = -y : x + y - 1$; and consequently (L, M) are connected by a quadric relation $(*L, 1)^2 = 0$, that is, the line envelopes a conic.

The locus, $(F, P, Q, R = 0$, where symmetrical or not, leads us to the following theorem of X, Y are quartic functions of a respectively; viz. these are unlike as like functions of the two variables according as the integral equation is not or is symmetrical. In the present case, however, where X is supposed to be the same function of a variable, but does the equation be a connection. In every chord through any two points of the conic, then the relation is obviously

$$\frac{dx}{\sqrt{(X)}} \pm \frac{dy}{\sqrt{(Y)}} = 0$$

where X, Y are quartic functions of x, y respectively, viz. this is also the symmetric relation. In the case of the in-and-circumscribed polygon there is an unsymmetric relation, but I cannot, however, propose to recall the integral equation in this case.

Attending now to the symmetrical case; if A and B are corresponding points, then the corresponding points of B are A and a new point C ; those of C are B and a new point D , and so on; so that the points form a series $A, B, C, D, \dots$: and the polynomial of A is, if a given position of B this series closes at a certain term, for instance, if $D = A$, then it will always thus close; whatever the position of the point on the conic, the integral equation

$$\Pi(y) - \Pi(x) = 0,$$

this equation must be a transformation of the original equation $(\ast \check{a} x, 1 \check{a} y, 1)^2 = 0$, and equally with it represent the relation between the parameters x, y of the two points A, B ; the constant of integration k of course be determined according as the coefficients of the form of the unsymmetrical case is so on.

Hence forming the equations for the correspondences $B, C; C, D; \dots$, and assuming that the series closes at $D = A$, we have

$$\Pi(y) - \Pi(x) = \Pi(z),$$

$$\Pi(z) - \Pi(y) = \Pi(z),$$

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